THE STEINBERG RELATION FOR THE SYMBOL OF A ONE-DIMENSIONAL MODULE OVER A NOETHERIAN LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02Q8 (sources/stacks-chow/pinned-original-source.tex line 17660).

Allowed standard prerequisites (editorial).

- The symbol d_M(a, b) of a pair of elements on a one-dimensional module and its multiplicativity. Same chapter, Lemma labelled lemma-compute-symbol-M and Lemma labelled lemma-symbol-compare-modules, cited at proof lines 17672-17673 and 17716-17718, with the relative symbols of Lemma labelled lemma-pre-symbol at proof lines 17705-17709. Supplies the object whose relation is asserted, the reduction to prime cyclic modules, and the comparison between relative and absolute symbols.
- Identification of determinants over A and over a quotient A/q. Same chapter, Lemma labelled lemma-determinant-quotient-ring, cited at proof lines 17683-17685. Justifies replacing A by the one-dimensional domain A/q.
- Two-element factorizations in a one-dimensional local Noetherian domain. Same chapter, Lemma labelled lemma-Noetherian-domain-dim-1-two-elements, cited at proof lines 17693-17695. Produces the extension A' and the factorizations a = t a', b = t b' with (a', b') = A'.
- Periodic determinant computations and the sign of d^A_{A'}(b', b'). Same chapter, Lemmas labelled lemma-periodic-determinant-easy-case, lemma-periodic-determinant and lemma-periodic-determinant-sign, cited at proof lines 17736-17758. Evaluates the three relative symbols through the displayed decompositions and supplies the final sign.

Source reference. Original lemma 02Q5: sources/stacks-chow/pinned-original-source.tex, source lines 17577–17577. Same-chapter pointer into the author environment lemma at source line 17577; invoked at proof line 17673 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02PC: sources/stacks-chow/pinned-original-source.tex, source lines 16434–16434. Same-chapter pointer into the author environment lemma at source line 16434; invoked at proof line 17684 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02Q2: sources/stacks-chow/pinned-original-source.tex, source lines 17483–17483. Same-chapter pointer into the author environment lemma at source line 17483; invoked at proof line 17689 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AY9: sources/stacks-chow/pinned-original-source.tex, source lines 17472–17472. Same-chapter pointer into the author environment lemma at source line 17472; invoked at proof line 17690 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02Q7: sources/stacks-chow/pinned-original-source.tex, source lines 566–566. Same-chapter pointer into the author environment lemma at source line 566; invoked at proof line 17695 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02PX: sources/stacks-chow/pinned-original-source.tex, source lines 17371–17371. Same-chapter pointer into the author environment lemma at source line 17371; invoked at proof line 17706 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02Q4: sources/stacks-chow/pinned-original-source.tex, source lines 17533–17533. Same-chapter pointer into the author environment lemma at source line 17533; invoked at proof line 17710 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02PN: sources/stacks-chow/pinned-original-source.tex, source lines 16696–16696. Same-chapter pointer into the author environment lemma at source line 16696; invoked at proof line 17731 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02PO: sources/stacks-chow/pinned-original-source.tex, source lines 16730–16730. Same-chapter pointer into the author environment lemma at source line 16730; invoked at proof line 17732 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02PM: sources/stacks-chow/pinned-original-source.tex, source lines 16676–16676. Same-chapter pointer into the author environment lemma at source line 16676; invoked at proof line 17755 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let A be a Noetherian local ring. Let $a, b \in A$. Let M be a finite A-module of dimension 1 on which each of a, b, $b - a$ are nonzerodivisors. Then*

$$d_M(a, b-a)d_M(b, b) = d_M(b, b-a)d_M(a, b)$$

in κ^*.

Proof. By Lemma 02Q5 it suffices to show the relation when $M = A/\mathfrak{q}$ for some prime $\mathfrak{q} \subset A$ with $\dim(A/\mathfrak{q}) = 1$.

In case $M = A/\mathfrak{q}$ we may replace A by $A/\mathfrak{q}$ and a, b by their images in $A/\mathfrak{q}$. Hence we may assume $A = M$ and A a local Noetherian domain of dimension 1. The reason is that the residue field κ of A and $A/\mathfrak{q}$ are the same and that for any $A/\mathfrak{q}$-module M the determinant taken over A or over $A/\mathfrak{q}$ are canonically identified. See Lemma 02PC.

It suffices to show the relation when both a, b are in the maximal ideal. Namely, the case where one or both are units follows from Lemmas 02Q2 and 0AY9.

Choose an extension $A \subset A'$ and factorizations $a = ta'$, $b = tb'$ as in Lemma 02Q7. Note that also $b - a = t(b' - a')$ and that $A' = (a', b') = (a', b' - a') = (b' - a', b')$. Here and in the following we think of A' as an A-module and a, b, a', b', t as A-module endomorphisms of A'. We will use the notation $d^A_{A'}(a', b')$ and so on to indicate

$$d^A_{A'}(a', b') = \det_\kappa(A'/a'b'A', a', b')$$

which is defined by Lemma 02PX. The upper index ^{A} is used to distinguish this from the already defined symbol $d_{A'}(a', b')$ which is different (for example because it has values in the residue field of A' which may be different from κ). By Lemma 02Q4 we see that $d_A(a, b) = d^A_{A'}(a, b)$, and similarly for the other combinations. Using this and multiplicativity we see that it suffices to prove

$$d^A_{A'}(a', b' - a')d^A_{A'}(b', b') = d^A_{A'}(b', b' - a')d^A_{A'}(a', b')$$

Now, since $(a', b') = A'$ and so on we have

$$\begin{array}{ccc} A'/(a'(b' - a')) & \cong & A'/(a') \oplus A'/(b' - a') \\ A'/(b'(b' - a')) & \cong & A'/(b') \oplus A'/(b' - a') \\ A'/(a'b') & \cong & A'/(a') \oplus A'/(b') \end{array}$$

Moreover, note that multiplication by $b' - a'$ on $A/(a')$ is equal to multiplication by b', and that multiplication by $b' - a'$ on $A/(b')$ is equal to multiplication by $-a'$. Using Lemmas 02PN and 02PO we conclude

$$\begin{array}{ccc} d^A_{A'}(a', b' - a') & = & \det_\kappa(b'|_{A'/(a')})^{-1} \det_\kappa(a'|_{A'/(b'-a')}) \\ d^A_{A'}(b', b' - a') & = & \det_\kappa(-a'|_{A'/(b')})^{-1} \det_\kappa(b'|_{A'/(b'-a')}) \\ d^A_{A'}(a', b') & = & \det_\kappa(b'|_{A'/(a')})^{-1} \det_\kappa(a'|_{A'/(b')}) \end{array}$$

Hence we conclude that

$$(-1)^{\text{length}_A(A'/(b'))} d^A_{A'}(a', b' - a') = d^A_{A'}(b', b' - a')d^A_{A'}(a', b')$$

the sign coming from the $-a'$ in the second equality above. On the other hand, by Lemma 02PM we have $d^A_{A'}(b', b') = (-1)^{\text{length}_A(A'/(b'))}$ and the lemma is proved. □